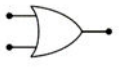
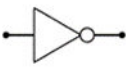
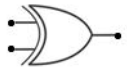


3.4.2 Boolean logic

Content	Additional information
Construct truth tables for the following logic gates: • NOT • AND • OR • XOR.	Students do not need to know about or use NAND and NOR logic gates.
Construct truth tables for simple logic circuits using combinations of NOT, AND, OR and XOR gates. Interpret the results of simple truth tables.	Students should be able to construct truth tables which contain up to three inputs.

Content	Additional information
Create, modify and interpret simple logic circuit diagrams. Students will only need to use NOT, AND, OR and XOR gates within logic circuits. Students will be expected to understand and use the following logic circuit symbols:  AND  OR  NOT  XOR	Students should be able to construct simple logic circuit diagrams which contain up to three inputs.
Create and interpret simple Boolean expressions made up of NOT, AND, OR and XOR operations.	Students will be expected to understand and use the following Boolean expression operators: . to represent the AND gate + to represent the OR gate $\oplus$ to represent the XOR gate - Overbar to represent the NOT gate For example the expression (A AND B) OR (NOT C) would be represented as: $(A \cdot B) + \bar{C}$
Create the Boolean expression for a simple logic circuit. Create a logic circuit from a simple Boolean expression.	